



**WHEN SCHEDULE
OVERRULED SAFETY:
A PROJECT MANAGEMENT
ANALYSIS OF THE SPACE
SHUTTLE CHALLENGER
DISASTER**

Project Management

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SUMMARY

One of the most notorious failures of a project in recent American history was the Space Shuttle Challenger tragedy. The accident happened on January 28, 1986, when mission STS-51L departed from Kennedy Space Center during the twenty-fifth launch of the Space Shuttle. After only 73 seconds into flight, the Space Shuttle exploded, and all seven astronauts died as a result. This incident could technically be attributed to the malfunction of an O-ring at the right solid rocket motor of Challenger under exceptionally cold pre-launch conditions. While many people perceive this as an engineering fiasco, it also needs to be considered a project management disaster since a mission that had an objective, deadline, resources, contractors, stakeholders, quality and safety standards, and desired result did not work out due to failure of the project system.

In this study, it will be explained that the Challenger accident occurred due to the decision makers viewing the decision of being ready to launch as a milestone based on the calendar and not on a decision for value on safety issues. The Shuttle, the people onboard, the contractors, the launch site, the weather conditions, the spectator audience, and NASA itself are part of one interconnected project system. Within such a system, there had been many indications before making the launch decision that it was best to postpone. Nevertheless, there was no action taken by the organization to use these indications and make a no-go decision. The example highlights how a project, which is technically complex, might fail in case there is a lack of synergy between its different aspects. [Appendix A](#) maps these ideas to the major project management concepts from the PMBOK Guide (7th Edition) used throughout the report.

This example of the Challenger disaster is particularly relevant in demonstrating the point that project success does not equal visible effort. The case has direct relevance to value creation, project context assessment, stakeholder management, creation of a collaborative culture, process adaptation to the high-stakes environment, indicator measurement, and uncertainty handling. This example further demonstrates that leadership is different from authority. While authorities can make decisions regarding the approval for launching a spacecraft, leadership requires a decision-making process aimed at determining if the launch ought to be made. For a project involving human life, the desired project success will not involve only completing the visible task.¹

In this regard, the mission served as a good illustration in that it highlights the gap between procedure and productivity. NASA had its readiness reviews, papers, contractors, organizational hierarchy, and technical experts. On paper, everything appeared organized. The problem was that there

¹ Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK Guide) and The standard for project management (7th ed.). Project Management Institute.



was no guarantee that the right safety information remained salient. Effective project management is not just about creating schedules, reports, and signatures. It provides safeguards against the illusion of control through data, disagreement, power to make decisions, and accountability. In the Challenger case, those safeguards were at their weakest when they should have been strongest. From a value perspective, the mission was to provide scientific, educational, national, and organizational benefits. The obvious outcome was the launch. The deeper outcome was a safe spaceflight mission that would achieve its objectives and return the crew safely. This is important because a project may appear to be successful when it delivers an output, but fails when that output does not create the intended value. Challenger teaches us that activity completion and value realization are different.²

FAILED PROJECT ANALYSIS

Project Background and Value

STS-51L was not a normal operational task. It was a temporary effort put together to achieve a unique result. It required a crew, payloads, hardware, contractors, mission control, launch operations, public communication, and several levels of readiness review. The mission also had symbolic importance, because it was carrying Christa McAuliffe, the first participant in NASA's Teacher in Space Project. This raised the profile of the mission in the public's eye and in schools around the country, but visibility should have increased the responsibility to handle the project with care, not diminishing the tolerance for delay.

From a value perspective, the mission was to provide scientific, educational, national, and organizational benefits. The obvious outcome was the launch. The deeper outcome was a safe spaceflight mission that would achieve its objectives and return the crew safely. This is important because a project may appear to be successful when it delivers an output, but fails when that output does not create the intended value. Challenger teaches us that activity completion and value realization are different.

The project failed because one project output, the launch, was allowed to dominate the larger project outcome, safe mission completion. A launch on time is valuable only if it supports the larger mission. Once the launch destroyed the vehicle and crew, the apparent schedule achievement became meaningless. This is why the case fits PMBOK Section 3.4, Focus on Value. Value is not measured by

²Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK Guide) and The standard for project management (7th ed.). Project Management Institute.



how fast an activity is completed. It is measured by whether the project outcome creates the intended benefit for the stakeholders who depend on it. In Challenger, the value chain broke at liftoff.

The value of the project also depended on trust. NASA was not just running a spacecraft; it was running public confidence in a national space program. The shuttle program was founded on the concept that a reusable spacecraft could be operated safely repeatedly. The Challenger accident damaged that trust because the public learned that the disaster was not a random surprise. The solid rocket booster joints and O-rings were known to be a concern before launch. No project can be divorced from the handling of a safety-critical weakness once there is evidence of a weakness.

[Appendix B](#) shows how the project moved from earlier O-ring concerns to the final launch decision. The timeline is important because it shows that the final failure was not isolated to one morning. A project can fail gradually before it fails visibly. Each earlier anomaly, each unresolved question, and each missed escalation became part of the outcome. The launch was the moment of disaster, but the project management failure developed over time.

The Project Environment

The decision to launch the Challenger was influenced by both the internal and external project environments. Inside NASA, formal reviews were conducted on readiness, levels of management, contractor certifications, engineering teams, launch criteria, and historical data from previous flights. From the outside, the mission was dictated by the weather, what people wanted to see, the press, political spotlight, and the general pressure to show that shuttle launches could be routine. Not all these environmental factors were equal. The most proximate external influence was the environment – the freezing weather at Kennedy Space Center before launch.

According to the Rogers Commission, the temperature at launch was 36 degrees Fahrenheit, far colder than previous launches. The Commission estimated the temperature at the coldest point on the right aft field joint close to the location of the failure to be 28 degrees Fahrenheit, plus or minus 5 degrees. That was important because the joint seal relied on the O-rings to respond quickly at ignition. The O-rings were cold and less springy. In other words, environmental temperature change caused a safety-critical component to behave differently. A good project process would have recognized this environmental condition as a major launch constraint.

This external situation did not call forth a sufficient response from the internal environment. The project had formal processes, but formal processes are only useful if they surface the right information to the right decision makers. The Commission found fault with the decision to launch, in part because the



decision makers lacked an understanding of the recent history of O-ring problems, the contractor's initial recommendation not to launch below 53 degrees Fahrenheit, or the continued opposition of some Thiokol engineers after management changed its position.³ Here is where it started to get dangerous in the project environment: The technical environment was uncertain, and the organizational environment filtered and softened that uncertainty.

This section directly correlates with PMBOK section 2.4: The Project Environment. A project does not exist in isolation. It is affected by internal factors like culture, governance, documentation, availability of resources, and the capability of the employees. It is also influenced by external factors such as weather, regulation, industry standards, social expectations, and public visibility. In Challenger, the physical environment was an immediate threat to safety, but the internal culture and governance environment did not respond proportionately. Both environments were to be handled together.

It wasn't simply a situation that existed; rather, it was an operational context shift. What worked in a warm climate may not work in a cooler one if component behavior is altered. This is precisely why a process needs to be customized to fit the existing context rather than the ideal or normal one. The question shouldn't have been: "Can we stick with our scheduled launch?" Rather, the question was: "Does the project still have the criteria required for launch?" The answer was far from clear.

Another environmental factor that could have caused a problem was the interaction between NASA and its contractors. Morton Thiokol possessed technical expertise in the solid rocket booster, while NASA held program authority. This would complicate decision-making because the contractor must express his concerns about the safety of the O-rings, yet at the same time must preserve its relationship with the client. On the other hand, NASA must solicit the opinions of the contractor while at the same time managing its program schedule. Under normal project conditions, the issues mentioned above would be anticipated and managed accordingly.

Stakeholders and Team Communication

The management of stakeholder knowledge was one of the biggest problems of the Challenger project. Stakeholders were NASA executives, launch directors, astronauts, astronauts' family members, engineers, and managers at Thiokol, Rockwell International, the government of the United States, the students who watched the Teacher in Space program, and the citizens of the country. All these

³ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Report to the President: Volume 1, Chapter V: The contributing cause of the accident. NASA History Office. The Commission found that the launch decision was flawed and that key decision makers lacked important information about O-ring history and Thiokol's recommendation.



stakeholders had different levels of power; however, some of them had relevant knowledge for the project's success. For instance, the Thiokol engineers were high-value stakeholders because they knew more about the danger that could arise from the O-rings than other persons responsible for the decision to launch the space shuttle⁴.

The dissenting voice of the technical personnel in the project team was not wholly protected during this process. Prior to the launch of the Challenger, during the night teleconference, the staff of Thiokol presented concerns relating to the low temperatures and O-ring sealing. The initial recommendation from an engineering standpoint was not to launch under the previously experienced temperature of 53 degrees Fahrenheit.⁵ This was not a minor issue. Rather, it was the threshold set by limited data. While the engineers could not determine precisely the point of failure for the joint, their inability to measure the risks should have heightened their caution.

This is an example of a communication problem as well as a problem of decision-making. For a risky venture, the burden of proof should not be on the presence of a risk but on ensuring safety. If a certain part of a project has not been proven to function safely under its expected conditions, then the project must be halted until more information is provided. Challenger provides an example of what results from the confusion between uncertainty and lack of proof. Uncertainty does not imply lower risks; it means higher uncertainty.

It is apparent from the above that the team culture did not support constructive conflict either. A good project team recognizes that it must generate conflict to identify risks before they become liabilities. However, in this case, there was a conflict between engineering facts and management opinion, but no proper safety process was in place to solve the issue. It was reported by the Commission that there were breakdowns in communications and the provision of accurate information, and that there was an organizational structure within NASA that enabled flight safety problems to elude shuttle managers.⁶

This is connected to PMBOK Section 3.2, Create a Collaborative Project Team Environment, and the Team Performance Domain. Collaboration does not mean agreement for the sake of agreement. It

⁴ Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK Guide) and The standard for project management (7th ed.). Project Management Institute.

⁵ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Report to the President: Volume 1, Chapter V: The contributing cause of the accident. NASA History Office. The Commission found that the launch decision was flawed and that key decision makers lacked important information about O-ring history and Thiokol's recommendation.

⁶ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Chapter V. Used here for the Commission's findings about communication failures, the silent readiness review record, and management isolation.



means creating a culture where people can raise concerns, challenge assumptions, and participate in decisions that affect project outcomes. A collaborative environment requires role clarity, open communication, trust, and respect. In Challenger, technical disagreement existed, but the decision system did not keep that disagreement visible enough. The appearance of consensus became more important than the reality of unresolved concern.

The most important lesson that I drew from the exercise is that teams on projects require an avenue whereby negative news can be safely communicated. Communication of positive news is normally straightforward, as positive news helps to maintain momentum in projects. Negative news causes delays and conflicts and creates issues of accountability. Projects facing high risks must develop a mechanism for communicating bad news. When engineers must temper their message to communicate bad news, part of the safety mechanism has been bypassed.

Planning and Launch Readiness

Planning is more than just creating an agenda or a schedule. Planning involves the coordination of preparation that will allow for the achievement of the project objectives under real circumstances. In relation to the Challenger disaster, the process of launch readiness could not adjust to fit the current environment. A readiness review will ensure that the vehicle has been made ready based on certain anticipated environmental conditions. If there are changes to these environmental conditions, the readiness plan must be reconsidered.

The PMBOK Planning Performance Domain would be helpful in the above situation since it views planning as not merely an exercise in creating one-off documentation. The planning process involves elements like scope, schedule, resource management, communication, procurement, change, metrics, and alignment. Challenger's planning for readiness required that all these elements be accounted for. The launch schedule, contractor input, launch conditions, hardware testing, crew safety, and management approval had to come together. Any alteration in the key variables required a reconsideration of the entire plan. The low temperature was not just another factor.

There was a need for a proper understanding of what it meant. In relation to STS-51L, done was not supposed to imply that everything was ready since it included completion of documentation, counting down to zero, and having an open launch window. Rather, done was supposed to imply that everything was in sync, including the shuttle, launch environment, certification from the contractor, crew safety assumptions, and any critical technical risk areas. Any deviation from the known limits of any one safety-critical item would revert readiness to the review stage, hence delaying launch.



The Flight Readiness Review process was created to bring issues up, but in this situation, it failed to do so. The report revealed that the issue regarding the O-rings was not addressed through the flight readiness process, and in five inches worth of documents about the 51-L readiness review, there was no reference made to the issues at all.⁷ Thus, while the process existed, it failed in the very purpose it was meant to serve.

The other planning problem was the lack of cold-weather guidelines. If a critical piece of safety equipment had never been tested under a certain condition, then the launch process would recognize this condition as falling outside the established envelope. The problem with the Challenger program was the project team's trying to decipher what the incomplete information meant on the eve of a launch. This is not how a high-stakes project should go about things. Acceptance criteria must not be made up at such meetings; they need to be known ahead of time.

Trade-offs are another aspect of planning that needs to be managed effectively. If a project encounters an issue involving the project's schedule, budget, quality, and safety, some element will have to yield to others. In a typical business-related project, for instance, a project manager can decide to allocate more resources, lengthen the project schedule, decrease the project's scope, or make a trade-off in terms of the project's quality. On the other hand, when a project involves sending a human crew to space, safety should outweigh the schedule.

Quality and Delivery Failure

The actual technical failure occurred in the area where the joint connecting sections of the right solid rocket booster met. O-rings were supposed to stop any leakage of combustion gases out of the joint during motor firing. The findings of the Rogers Commission indicated that there was a combustion gas leak from the aft field joint of the right solid rocket motor that started just after ignition and continued until the weakening or puncture of the external fuel tank destroyed the spacecraft.⁸ While this means the O-rings are a point of technical failure, the larger issue of quality failure extended beyond the rubber components or joints.

The concept of quality, in this scenario, had multiple facets. For instance, the part needed to be reliable in the conditions where it would be expected to operate during the launch. The next facet

⁷ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Chapter V. The Commission found that no mention of the O-ring problems appeared in several inches of Flight Readiness Review documentation for 51-L.

⁸ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Chapter IV. The Commission found that a combustion gas leak through the right Solid Rocket Motor aft field joint began at or shortly after ignition and led to vehicle breakup.



pertained to verification that the part in question was safe enough to use. Finally, there were certain acceptance criteria related to the launch itself that were needed to accurately reflect the level of danger. According to the Commission's research, the performance of the O-rings was affected by their temperature.⁹ If a seal needs to be quick at ignition time, its resiliency cannot be overlooked.

This relates to the PMBOK Guide, Section 3.8, Building Quality into Processes and Deliverables. Quality does not mean just that the end-product functions properly after delivery. Quality means that the processes involved in creating and approving the deliverable can produce an end-product that is safe and functional for its intended purpose. In the case of the Challenger, the deliverable did not include just a launching vessel; rather, it included safe mission capability.

This incident is a prime example of the need for quality control before production as well. The O-ring deterioration and blow-by in past launches were already signs of trouble. It was supposed to be a process defect that had to be corrected prior to the next time such conditions occurred during launch. Unfortunately, these defects continued until they were considered normal. Organizations tend to view any anomaly as acceptable if it did not cause a catastrophe on its first occurrence.

The failure in delivering value came because the mission did not get to a place where its benefits could have been reaped. The mission delivered movement, smoke, and lift-off, but not the results expected. This is why the case of Challenger can be used to provide an excellent illustration of a failed project. The launch was not everything when it came to deliverable value.

Systems Thinking and Complexity

The problem with Challenger was not the result of a single error. Rather, it was a systemic error that included the functioning of the hardware, the weather, communications, management structure, pressure from contractors, normalizing history, and public perception. Systems thinking is crucial because the launch vehicle itself was just one element of the whole project. The mission occurred within a larger system, including NASA centers, contractors, readiness checks, political pressure, funding issues, launch dates, and public relations.

This relates to PMBOK 3.5, Recognizing, Evaluating, and Responding to System Interactions. A project is a system composed of interrelated parts. The interaction between the technical system and the organizational system occurred. The effect of cold temperature on O-rings created uncertainty in O-rings.

⁹ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Chapter IV. The Commission found that O-ring resiliency is directly related to temperature and that a cold O-ring may not follow the opening tang-to-clevis gap fast enough to seal.



Uncertainty about O-rings led to recommendations made by the contractors. Recommendations made by contractors influenced NASA's decision to launch the rocket. Launching the rocket by NASA influenced the lives of astronauts and the reputation of the program.

The technical system in and of itself was complicated. The field joint of the solid rocket booster had to remain sealed during ignition, even as the various pieces of equipment were moving apart from one another and trying to release the gases that were generated. The organizational system made things even more complicated. Each party involved had a different piece of the puzzle when it came to the information available, as well as different motivations for what needed to happen next. The engineers were concerned about uncertainty and equipment.

Ambiguity became an aspect of complexity as well. The existing data did not entirely help predict what would occur at the anticipated launch temperature. The engineers knew that the launch was not within the safe experience base; however, the engineers could not provide the exact probabilities of failure. This uncertainty became a problem for management in their perspective on the decision to recommend against launching. What should have been asked instead was why there was any problem moving forward, given that the exact probability could not be determined when seven lives were at stake?

Section 3.9: Navigate Complexity from PMBOK is also applicable in this case due to the existence of human factors, systemic factors, uncertainties, ambiguities, and complexities of technology. The problem was not only the complexity of the shuttle itself but rather the complexity associated with the project, which involved many components, each of whose interaction made the total impact hard to understand. Technical issues became even more critical because of the organizational complexity surrounding it.

The tragedy illustrates another aspect in which a system might damage itself over time. Previous O-rings were not responsible for the destruction of the entire vehicle, and hence, the organization became used to flying even with erosions. According to the Report of the Presidential Commission on the Space Shuttle Challenger Accident, the solid rocket motor case joint was categorized as a Criticality 1 component, which implies that any failure here would lead to mission, vehicle, and crew loss.¹⁰ In the case of a functioning system with a Criticality 1 problem, the level of risk management must be extremely high.

¹⁰ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Chapter VI: An accident rooted in history. NASA History Office. The report describes the Solid Rocket Motor case joint as a Criticality Category 1 item whose failure could cause loss of mission, vehicle, and crew.



Risk, Measurement, and Uncertainty

The major problem involved with the tragedy of Challenger involves risk management. In this case, the project team knew there was a danger since the O-rings might fail to operate correctly at low temperatures. In this scenario, the risk management technique that was used was risk acceptance since the launch went ahead despite this danger. But it is important to note that risk acceptance was not done responsibly. There must be risk owner identification, stakeholder involvement, reasons stated, and consequences understood.

It is linked to PMBOK's Section 3.10 - Optimize Risk Responses. The risk responses must be commensurate with the level of the risk, cost-effective, realistic in the context of the project, acceptable to all the key stakeholders, and owned by someone responsible for the action. In the case of Challenger, the negative risk level was very high since its consequences were total loss of mission, aircraft, and crew. So, the risk response must have been conservative; that is, a delay in launching was proper.

It should have been done by avoiding the risks. Avoiding the risk is not synonymous with discontinuing the shuttle program. It involved getting rid of the cause of the risk by postponing the launch of the shuttle until such a time when the temperatures were within an acceptable range, or the joint was tested while it was still cold. It would have been the easiest and most ethical action to take.

Another problem with the project was the lack of useful leading indicators. A good measuring instrument should do more than tell us if we're on track to launch. It should tell us if our planned launch is even safe. Some leading indicators for us to consider would be O-ring damage experience, minimum predicted joint temperature, safety objection issues outstanding, being in a condition untested, and violating an engineering recommendation. All these indicators would have put the status of the launch on red before launch.¹¹ A dashboard mock-up for such a system can be found in [Appendix E](#).

Measurement in the Performance Domain of PMBOK is critical because it affects behavior. For instance, when the leadership focuses on measuring the frequency of launches rather than the safety level, the employees will realize that scheduling is what truly counts. When the leadership measures the existence of dissension, temperature margins, Critical 1 issues, and test-boundary breaches, then there is a possibility that safety might be addressed before it is too late.

But uncertainty was not just a matter of engineering. People had different ideas about the meaning of the data, the person responsible for making the decision, and the level to which the risk should

¹¹ Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK Guide) and The standard for project management (7th ed.).



be escalated. The Uncertainty Performance Domain can help us understand why this was the case because it considers risk, ambiguity, complexity, volatility, and the environment.¹² In this case, the challenge had everything to do with these concepts. The cold temperatures created uncertainty about the physical aspects of the mission. The O-ring data created technical ambiguity.

Feynman's appendix on the report of the Rogers Commission similarly identifies another issue regarding measurements. He noted that there was a large difference between the estimates of shuttle risk provided by engineers and those made by managers.¹³ Engineers were more pessimistic about the risk involved compared to managers. Such differences are indicative of potential problems. If the individuals who understand technology do not share the same perceptions as their leaders, then this discrepancy should be investigated rather than averaged out.

As depicted in [Appendix D](#), the main risks were not arbitrary. They were identifiable, such as O-rings working poorly in cold weather, miscommunication, normalization of deviance, pressure on the schedule, and lack of proof in real-world situations. The use of a risk register would not alone solve the problem; however, it would force the project team to write down the risk, give it an owner, specify a course of action, and tie it into a launch criterion. Just having the risk written down makes it difficult for the organization to ignore it.

Leadership, Governance, and Stewardship

The Challenger incident highlights the contrast between authority and leadership. The organization had its authority structures: readiness reviews, program level, management, contractor relations, and signatures. Authority alone, however, could not yield a safe choice. Leadership entailed slowing the whole process down, safeguarding dissent, seeking evidence of safety, and raising unresolved issues to higher levels. Leadership becomes particularly crucial when there is momentum in the project timetable.

Demonstrating Leadership Behaviors, PMBOK Chapter 3.6, is important because leadership does not come with a nameplate. It is the responsibility of the project leader to lead his or her team to the right answer rather than simply getting the job done. In the Challenger scenario, leadership would have entailed

¹² Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK Guide) and The standard for project management (7th ed.).

¹³ Feynman, R. P. (1986). Personal observations on reliability of shuttle. In Report of the Presidential Commission on the Space Shuttle Challenger Accident, Appendix F. NASA History Office. Feynman noted large differences between working engineers' and management's estimates of shuttle failure probability.



posing some difficult questions, such as: What is the opinion of those most knowledgeable about the risk? Are we operating within the limits of our qualified experience base? What happens if the dissenters are correct?

Stewardship was equally important. The leadership at NASA was stewards for the taxpayers' money, the trust of the nation, the relationship with contractors, the scientific goals, and the safety of the astronauts. According to a definition by Webster's Dictionary, a steward is someone who exercises diligence, integrity, responsibility, and compliance.¹⁴ In the case of Challenger, diligence would mean giving precedence to astronaut safety over deadlines. Integrity would be ensuring that contrary opinions on safety issues get enough prominence. Responsibility would mean never adopting a procedure that makes an engineer feel coerced into changing his opinion.

The problem of governance lies in the fact that the issue of safety was not brought to the proper level with the required gravity. According to the Commission's investigation, important information regarding previous incidents with O-rings and Thiokol engineers' warning had not reached Level I and II individuals, or the Launch Director, before the launch.¹⁵ It should be noted that it is not merely an issue of miscommunication. In fact, the problem of governance here lies in the disconnection between decision-making authority and safety information.

PMBOK Section 2.2 on organizational governance systems can also be applied here. Governance systems must ensure that there is an appropriate structure within which control, decision-making processes, problem-solving mechanisms, and risk assessments are conducted. There were some forms of governance systems in place for the Challenger case, but they did not facilitate a proper pathway for the dissemination of any safety information that remained unresolved. The governance system must ensure that it is difficult for any important safety information to become nonexistent.

Following the incident, the recommendations put forward by the Rogers Commission highlight the significance of quality and governance. They suggest redesigning the problematic solid rocket booster joint and seal, as well as an independent review of the redesign and certification process.¹⁶ The recommendations thus reveal that the issue was not limited to the defective part itself, but also to the

¹⁴ Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK Guide) and The standard for project management (7th ed.). Project Management Institute.

¹⁵ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Chapter V. The report states that crucial information about prior O-ring damage and Thiokol engineers' arguments did not reach key Level I and II program officials or the Launch Director.

¹⁶ Presidential Commission on the Space Shuttle Challenger Accident. (1986). Recommendations. NASA History Office. The Commission recommended redesigning the faulty Solid Rocket Motor joint and seal, independent oversight, improved communications, and stronger safety organization.



necessity of better independent review processes and safety mechanisms robust enough to withstand scheduling pressures.

[Appendix G](#) outlines the corrective actions that integrate leadership, governance, and stewardship. This is not a difficult concept: postpone launch, ensure safety certification, safeguard whistleblowing, split safety from time control authorities, apply leading indicators, consider anomalies as failures, customize controls to fit the situation, and properly define success. This is not a question of comprehending what those actions mean after something has gone wrong. This is a matter of having the leadership discipline to take those steps beforehand.

Safety Culture and Normalization of Deviance

Normalization of deviance happens when the organization accepts abnormality because the prior abnormality did not result in failure right away. The earlier erosion of O-rings would have signaled a major problem with the design and procedure. But because prior launches were successful, the abnormality was accepted as part of the operational experience. This is particularly risky in complicated systems. The system can withstand several warnings until the factors converge to eliminate all margins.

Psychological safety is also important for safety culture. Engineers and technical professionals need to feel that they can bring up their doubts without being labeled as a troublemaker, someone who lacks loyalty, or too paranoid. In a high-performing group, dissenting voices should be heard as dissent exposes the hidden danger. When the team considers dissidents as obstacles to meeting the deadline, it might fall silent. This silence might be interpreted as consent when the project is dangerous, although it could be a sign of fear, exhaustion, or resignation.

This is precisely the reason why [Appendices D](#) and [E](#) are relevant. Risk registers and ready dashboards are not mere pieces of paper. They are cultural tools that make the organization face facts and link them to decision-making. If the readied dashboard turns out to be red in terms of expected joint temperature, red regarding the history of O-rings, and red concerning outstanding contractor disagreements, then management cannot avoid making tough decisions, which would be either to abort the mission or to justify why not.

Corrective Project Management Lessons

The first corrective lesson is that high-risk endeavors should have specific no-go rules. The no-go rule is quite effective since it eliminates the burden placed upon an individual. It does not put the burden of convincing all the managers in the room on one engineer, but rather allows the endeavor to use a predetermined standard. If the situation is beyond the scope of the tests done, the endeavor ends. If there



is any disagreement over a Criticality 1 issue, the endeavor ends. If the safety data is incomplete, the endeavor ends.

Lesson number two is that the authority for safety should be separate enough to be able to overcome any scheduling pressure. However, this does not imply that the people who deal with safety issues should remain detached from the reality of the project itself. The authority should be strong enough for these people to be able to halt the operation in case of need. While a project manager should consider all constraints involved in the operation, some constraints carry greater importance than others.

The third lesson is for project teams to regard uncertainty as part of their decisions. Uncertainty is frequently regarded as a nuisance since it disrupts the flow of work. However, uncertainty is perhaps the most significant piece of information that the project team could possess. If a project team cannot determine whether a crucial element will work in practice, then it needs to be made evident through their decisions. This is not just based on the facts that indicate they should launch the product.

Lesson four is that context really matters. A shuttle launch does not have the same risk tolerance level as a simpler mission. Tailoring is stressed within the Project Management Body of Knowledge section 3.7, and Challenger is an ideal illustration of such necessity.¹⁷ In this case, there was a complex technical system, a large number of stakeholders, contractor influence, extremely serious consequences in case of failure, and even changing the nature of components due to environmental conditions.

CONCLUSION

The Challenger disaster was a failed project because it failed to deliver its intended outcome. The mission was not simply supposed to launch. It was supposed to safely carry a crew and payloads into space, complete mission objectives, and reinforce public confidence in the shuttle program. The project failed because a known safety risk was allowed to pass through a project system that should have stopped it.

The disaster is best understood as a combination of technical and managerial failure. The O-ring seal failed, but the launch decision process also failed. The cold weather mattered, but so did the way the organization responded to the cold weather. Engineering concern mattered, but so did whether that concern reached the final decision makers. Prior anomalies mattered, but so did the way they were

¹⁷ Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK Guide) and The standard for project management (7th ed.). Project Management Institute.



classified, reported, and normalized. Challenger therefore shows that project management is not paperwork around real work. In complex projects, project management is part of the safety system.

The strongest lesson is that high-risk projects must be managed around value, not momentum. Schedule adherence is important, but it cannot become the definition of success. When human life is involved, safety must be built into planning, quality, stakeholder engagement, measurement, risk response, and leadership. A delayed launch would have been disappointing. The actual launch was catastrophic. Challenger remains a reminder that uncertainty should never be hidden by confidence, and authority should never overpower evidence.

The PMBOK concepts used in this report help explain why the disaster was not just an engineering case. Value delivery shows why the launch itself was not enough. Stakeholder engagement shows why technical dissent had to be protected. Team performance shows why open communication mattered. Planning shows why changing conditions should have reopened the readiness review. Delivery and quality show why mission success requires safe performance, not just liftoff. Measurement and uncertainty show why the project needed better leading indicators. Leadership, stewardship, governance, complexity, risk, adaptability, and tailoring show why the project system had to respond before failure.

Challenger is still studied because the underlying project management issues remain relevant. Organizations still face schedule pressure. Teams still struggle to escalate bad news. Leaders still sometimes mistake confidence for evidence. Complex systems still produce warning signs that can be normalized. The case remains powerful because it shows the cost of getting those issues wrong. The project manager's job is to deliver. It is to make sure the project is worth delivering and safe enough to continue.

WHAT I WOULD HAVE DONE DIFFERENTLY

A table summarizing this section can be found in [Appendix G](#).

1. As someone responsible for the decision to launch the Challenger, the first thing I would do differently is postpone the launch altogether. The anticipated cold weather did not fit into the safety range of experience for O-ring seals. The first warning from the contractors about the inability to launch at temperatures less than 53 degrees Fahrenheit ought to have sufficed. This would be considered a risk avoidance tactic. The postponement would not have fixed all problems with the shuttle design program, but it would eliminate an urgent risk from cold O-rings.



2. It would be beneficial to develop a specific safety protocol whereby in any case where there are indications that conditions do not fall within the limits of anything that has been previously tested would be considered a no-go situation. In doing this, the burden of proof would shift from being required to prove disaster to having to prove safety. This is the only standard by which any human life-critical project should operate.
3. I would ensure that all safety-critical systems have been endorsed in writing by engineers when tested under actual launch conditions. Dissent would be recorded and escalated. Management could raise questions, but could not transform disagreement among engineers into authorization without further assessment. This would keep the project free from the influence of informal pressure, unclear consensus, and post hoc arguments regarding recommendations.
4. I would also ensure that the scheduling authority is distinct from the safety authority. The same hierarchy that oversees the scheduling process must not be the sole means of reaching an emergency stop decision when it comes to safety issues. There should be another group that is empowered to stop the mission when the threat is beyond the safety envelope. This does not mean that management will absolve itself of its duties, but that it will enhance its effectiveness through objective evaluation.
5. The launch readiness dashboard will incorporate leading indicators, which will include estimated joint temperature, minimum launch temperature recorded previously, history of O-ring erosion, any remaining criticality issues, pending contractor objections, and the degree of confidence in the safety assessment. Any red flag will necessitate action for resolution before the launch. The objective is not to generate unnecessary documentation. Instead, the objective is to ensure that the right information is discussed by the project team.
6. I would consider the occurrence of anomalies as higher risk and not lower. An erosion of the O-rings in previous launches should not have been normalized due to successful prior flights. It indicates that the margin is being eroded in a sophisticated safety system. Instead of launching into space, a proper investigation needs to take place, and redesigning is done to prevent any mishap. It would rather get criticism for being cautious instead of getting accolades for being reckless with the schedule.
7. The process for the project would need to be tailored to the specific environment. Shuttle launches are not ordinary operations. They are critical, technical, public, and involve human lives. This is not a routine process; therefore, it needs to be a more stringent one, with higher risk tolerances, stronger phase gates, mandatory escalation of objections, independent review, and clearly stated no-go criteria. This does not entail bureaucracy for the sake of bureaucracy; rather, it entails the appropriate amount of control in proportion to the potential peril involved.



8. I would include the astronauts and the safety officers in the final risk communication cycle if the criticality level 1 risk was still unresolved. That does not imply that the crew is expected to resolve every technical problem. It implies that the decision-making process should take into consideration the reality that the crew will carry the full burden of consequences. In case there is a dispute regarding an important safety issue that remains unresolved, the project cannot avoid responsibility using management lingo.
9. A post-anomaly resolution process must occur prior to the team accepting recurring problems as being normal business practices. The O-ring erosion anomaly would need to be analyzed, categorized, assigned to an individual, and closed when the root cause is determined and corrected, or else the risk remains visible on the readiness dashboard until the root cause is fixed. This prevents the team from gradually coming to accept abnormal performance as being normal performance.
10. I will redesign the launch decision meeting. Launch meetings do not require the engineering staff to justify why the launch would be unsafe. They need to determine if there is sufficient data to justify its safety. This kind of meeting can be structured so dissent comes first, along with recording uncertainties. Only after the safety review and management discussion does the process continue.
11. I would define project success differently. I would not define success as launching on time. I would define success as safely delivering the mission outcome. That definition would change every major decision. It would make delays acceptable when safety evidence is incomplete. It would make dissent valuable rather than inconvenient. It would make leadership responsible for asking uncomfortable questions before failure, not explaining failure afterward. In simple terms, I would rather defend a delayed launch than defend a preventable loss of crew and vehicle.



APPENDICES

Appendix A: PMBOK Application Map

This table shows exactly where the project management concepts are applied in the Challenger analysis.

PMBOK Area	How It Applies to Challenger	Best Use in Paper
2.1 Creating Value	The mission value was safe mission completion, not simply launching.	Project Background and Value
2.2 Organizational Governance Systems	Governance should connect decision authority with accurate safety knowledge.	Leadership, Governance, and Stewardship
2.3 Functions Associated with Projects	Oversight, resources, technical expertise, business direction, and governance had to work together.	Leadership and Planning
2.4 Project Environment	Cold weather, public visibility, NASA culture, and contractor relationships shaped the decision.	The Project Environment
3.1 Stewardship	NASA leaders were responsible for care, integrity, public trust, and crew safety.	Leadership and Governance
3.2 Collaborative Project Team Environment	The team needed open disagreement and safety-focused communication norms.	Stakeholders and Team Communication



3.3 Effectively Engage with Stakeholders	Safety-critical engineering concerns needed direct escalation and attention.	Stakeholders and Team Communication
3.4 Focus on Value	Launch schedule was confused with mission value.	Project Background and Value
3.5 Systems Thinking	Hardware, weather, contractors, management, and public pressure interacted.	Systems Thinking and Complexity
3.6 Leadership Behaviors	Leadership should have protected dissent and slowed the decision.	Leadership and Governance
3.7 Tailor Based on Context	A crewed spaceflight launch needed stricter safety gates and risk thresholds.	Corrective Lessons / What I Would Do Differently
3.8 Build Quality into Processes and Deliverables	The seal design and launch-readiness process did not prove safety under actual conditions.	Quality and Delivery Failure
3.9 Navigate Complexity	The disaster emerged from human, technical, and organizational complexity.	Systems Thinking and Complexity
3.10 Optimize Risk Responses	The proper response was risk avoidance through launch delay.	Risk, Measurement, and Uncertainty
3.11 Adaptability and Resiliency	NASA should have adapted to cold-weather evidence instead of holding the schedule.	Planning and Launch Readiness



3.12 Enable Change	Corrective action required changing safety governance and launch decision behavior.	Corrective Lessons
2.7 Measurement Performance Domain	Leading indicators should have flagged O-ring temperature and unresolved dissent.	Risk, Measurement, and Uncertainty
2.8 Uncertainty Performance Domain	Ambiguity, volatility, risk, and complexity shaped the launch decision.	Risk, Measurement, and Uncertainty
4.6 Artifacts	Risk registers, dashboards, stakeholder registers, and escalation models help preserve decision quality.	Appendices

Appendix B: Challenger Timeline

This timeline summarizes the most important project events and why each event matters from a project management perspective.

Date / Time	Event	Project Management Meaning
Before January 1986	Prior flights showed O-ring erosion and blow-by concerns.	Recurring anomalies should have increased risk urgency, not reduced concern.
January 15, 1986	51-L readiness review documentation did not clearly elevate O-ring concerns.	The readiness process existed but did not expose the safety-critical issue.



January 27, 1986, afternoon	Low-temperature concerns were raised by Morton Thiokol engineers.	External environmental conditions changed the launch readiness assumption.
January 27, 1986, evening	Thiokol engineers initially recommended against launching below 53 degrees Fahrenheit.	The project had a clear no-go signal tied to the available experience base.
January 27, 1986, later evening	Thiokol management reversed its position after internal caucus and recommended launch.	Governance allowed management judgment to override unresolved engineering dissent.
January 28, 1986	Challenger launched in unusually cold conditions and broke apart 73 seconds after liftoff.	The project delivered the visible launch but failed the intended mission outcome.
After the accident	The Rogers Commission investigated the failure and recommended redesign, independent oversight, improved communication, and stronger safety functions.	Corrective actions focused on design assurance, governance, communication, and safety authority.

Appendix C: Stakeholder Register

This register identifies the major stakeholder groups, and the communication need each group should have had in a safety-critical project.

Stakeholder	Primary Interest	Influence / Power	Communication Need
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Astronaut crew and families	Safe mission completion and transparent risk handling.	High moral importance; limited launch-decision power.	Direct awareness of critical launch risks and safety constraints.
NASA senior leadership	Mission success, program credibility, national confidence, and schedule performance.	High authority over program direction and launch readiness.	Clear escalation of unresolved safety-critical issues.
Mission Management / Launch leadership	Launch decision, countdown status, weather, and readiness closure.	High authority over final launch process.	Red/yellow/green safety indicators and dissent status.
Morton Thiokol engineers	O-ring performance, booster joint safety, technical integrity.	High technical knowledge; lower decision authority.	Protected channel for written dissent and no-go recommendations.
Morton Thiokol management	Contract performance, customer relationship, engineering certification.	Medium to high influence over contractor recommendation.	Clear separation between technical safety signoff and business pressure.
Rockwell International	Orbiter safety and ice/debris concerns.	Medium technical influence.	Clear process for communicating ambiguous safety concerns.
Students and public audience	Educational values, public trust, national pride.	Low direct power but high reputational impact.	Accurate public communication after decisions are made.



U.S. government / program sponsors	Space program performance, public accountability, and funding stewardship.	High strategic influence.	Evidence that safety governance is independent and credible.
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Appendix D: Risk Register

This register converts the main Challenger warning signs into a project risk format. The highest-priority response is risk avoidance through launch delay.

Risk	Warning Sign	Impact	Recommended Response	Owner
O-ring seal failure in cold temperature	Launch temperature below prior experience; known O-ring erosion history.	Loss of mission, vehicle, and crew.	Avoid: delay launch until temperature is within verified safe range or independent review clears launch.	Independent Safety Authority
Communication failure between engineering and launch leadership	Engineering dissent not escalated to final decision makers.	Launch based on incomplete information.	Mitigate: mandatory written dissent escalation and no-go review.	Program Governance Lead
Normalization of prior anomalies	Repeated O-ring erosion treated as acceptable because earlier flights survived.	Critical risk underestimated over time.	Mitigate: classify recurring erosion as a critical anomaly until root cause is resolved.	Quality / Risk Lead



Schedule pressure overriding safety logic	Launch was treated as a milestone rather than a value/safety decision.	Safety margins reduced to protect cadence.	Avoid/Mitigate: independent no-go authority and red readiness indicators.	Program Sponsor / Safety Office
Weak safety evidence under actual launch conditions	Cold-weather data is incomplete and not fully representative of flight configuration.	False confidence in component performance.	Mitigate: require verification under actual operating range before launch.	Engineering Review Board

Appendix E: Launch Readiness Dashboard Mockup

This dashboard mockup shows the kind of leading indicators that should have made the launch status red before liftoff.

Indicator	Status on 51-L	Meaning	Launch Rule
Predicted / actual joint temperature	RED	Below prior demonstrated launch experience and near freezing conditions.	No-go until within verified range or independently cleared.
Prior O-ring erosion / blow-by history	RED	Recurring anomaly on a Criticality 1 system.	No-go until root cause and corrective action are closed.
Contractor engineering signoff	RED	Initial engineering recommendation	No-go if technical dissent remains unresolved.



		opposed launch below 53 degrees Fahrenheit.	
Management signoff	YELLOW	Management recommendations existed but were changed after pressure and internal caucus.	Requires independent review when technical and management views diverge.
Weather / ice concerns	YELLOW	Cold weather and ice created additional unknowns.	Requires safety review and crew awareness if condition affects launch risk.
Public / schedule pressure	YELLOW	Visibility increased pressure to proceed.	Must not override safety gate criteria.
Final launch readiness	RED	Multiple unresolved red/yellow indicators remained.	Launch should be delayed.

Appendix F: Governance and Escalation Model

This model shows how unresolved safety dissent should move through a launch-governance system before a final launch decision.

Step	Governance Control	Purpose
1	Engineers identifies safety-critical concern under actual launch conditions.	Capture risk at the source before it is filtered by schedule pressure.



2	Concern is written in a safety dissent log with evidence, uncertainty, and recommendation.	Make dissent visible, traceable, and difficult to reinterpret later.
3	Independent Safety Authority reviews dissent and classifies severity.	Separate safety judgment from schedule and customer pressure.
4	If condition is outside tested or demonstrated limits, default status becomes no-go.	Shift burden of proof from proving danger to proving safety.
5	Program leadership can only override after documented independent review and crew-safety briefing.	Keep decision authority accountable and informed.
6	All open Criticality 1 concerns appear on the launch readiness dashboard.	Prevent unresolved critical risk from being hidden in lower-level records.

Appendix G: Recommended Corrective Action Checklist

This checklist summarizes what I would have changed if I were managing the launch-readiness decision.

Corrective Action	Reason	Expected Benefit
Delay launches under cold conditions	O-rings were outside the safe experience base.	Avoids the immediate triggering condition.



Create automatic no-go rules for unverified crewed-launch conditions	Human-life-critical projects require proof of safety.	Reduces pressure on individual engineers.
Require written engineering signoff and dissent escalation	Verbal or filtered communication can hide risk.	Improves traceability and accountability.
Separate schedule authority from safety authority	Launch can conflict with risk control.	Creates independent stop power.
Use leading safety indicators	Schedule metrics alone do not show readiness.	Makes unresolved risk visible before launch.
Treat repeated anomalies as increasing risk	Survival of previous flights does not prove safety.	Prevents normalization of deviance.
Tailor process to high-risk context	A crew launch requires stricter controls than a normal project.	Matches governance intensity to danger level.
Define success as safe mission completion	Launch is only an output, not the full value.	Aligns all decisions to the real outcome.



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